From business question to application

Homework: steps 1 to 3 from lecture 2 '*Asking the right questions*'

Group	F2			
Dataset / case	Project Food			
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Before we start building the dashboard, it is important that we know what to build. Fill In the following template: one per group. Take enough time to ensure you understand which questions to ask: you can make the world's most fancy dashboard, if it answers the wrong question, nobody will actually use it.

Step 1 · Understand the business question

Goal: one paragraph with the business problem, the decision it supports and who decides.

Business questions

Ask more than "what do you want to see?". Write your own open-ended question per topic, then fill in what the business owner would answer. Not all of these topics will be equally important. If you really can't think of a good question, then maybe that is a sign that this particular topic is not very relevant for your case. On the other hand, if you have many questions for a particular topic, you know how to pay more attention to it. Writing 'not relevant' for a topic is also a good answer, as is writing multiple questions.

Topic	Your question	Your answer
Problem *What is going wrong or unknown? Why now? What does the business owner want to solve?*	Why do dieticians need a tool to catch duplicate products and why?	Check if new products get scanned and added daily or weekly. Before a dietitian marks them as validated, someone must check whether the product is a duplicate of an existing validated product
Decision *What will someone do differently after using the dashboard?*	What do dieticians do differently once they have this tool?	The dietitian decides: • **Duplicate** → Do not validate as new; copy/merge the nutritional values form the matching active product. • **Not Duplicate** → Validate it as a new product and add missing nutritional values The developer decides which similarity methos and threshold work best for the textual and numerical datasets
People *Who will use it? Who else sees the result?*	Do the people who go to the dietician need to be able to see the dashboard?	• **Primary User:** Dietitians who validate new products. • **Secondary Users:** App developers, Data Scientists, Product Owner who maintain the similarity logic and dashboard

Workflow *How do you handle this today? Where do you get stuck? What could be better?*	How is this handled today and where does it break down?	• **Today:** Manual browsing or searching of the existing table for anything similar. If a new product is scanned, it'll be stored as active. If needed, the dietitian will have to fill in missing values • **Bottlenecks:** multilingual names, inconsistent brands/categories, multiple barcodes, missing nutritional values, and no systematic similarity ranking • **Better:** The app can suggest likely duplicate active products, shows evidence from text and numeric data, and proposes missing nutritional value with a reliable source
Rhythm *Daily, weekly, once per project? How often will people look at the dashboard?*	How often does someone actually use this?	On daily or per validation batch, depending on how many products are scanned.
Exploration *What do you compare, filter, zoom into? Which "what if" do you want to answer?*	What do they compare & what's the real "what if"?	• **Compare:** the new unvalidated product vs active validated products. • **What-if:** "What if I change the similarity threshold?", "What if I change the similarity threshold?" ⇨ **Textual and numerical datasets must be explored separately.**
Data *What data exists, at what detail, updated how often?*	What data exists, at what detail, updated how often?	The data that exists are: - **Categorical:** `name`, `name_search`, `active`, `remarks`, `brands`, `brands_search`, `bron`, `categories`, `barcode` - **Numerical:** `active`, `energy`, `protein`, `fat`, `saturated_fatty_acid`, `carbohydrates`, `sugar`, `starch`, `dietary_fiber`, `salt`, `sodium`, `categories` - **One row per product.** `active = 0` means unvalidated; `active = 1` means validated. ⇨ Missing values are common. Barcodes can contain multiple codes. Categories can contain multiple labels. The two datasets must **not** be merged.
Success *How will you know the dashboard helped?*	How well did this impact the users' goal?	Fewer duplicates in the active database, less time spent per validation, fewer missing nutritional values, faster backlog processing, and fewer incorrect analyses.

Summary

➔ What will be the main focus/problem that this dashboard answers?

[Client] needs to *[decision/action]* because *[problem]*. *[Users]* will use the app *[how often]* to *[main tasks]*, based on *[data]*.

The **app/developer/dietician** needs to **decide before validating each newly scanned product**, whether it is a duplicate of an existing active product or genuinely new, because **unvalidated products are accumulating and duplicates clutter the database**, leading to incorrect analyses and wasted dietitian time. **Dietitians** will use the app **during daily/weekly validation sessions** to **search, compare, confirm duplicates, and complete missing nutritional values** from reliable sources, based on **two separate datasets**: textual/categorical product data and numerical nutritional data.

Tip: if your summary only contains numbers to monitor ("see total sales"), you have a KPI dashboard. Find tasks: comparing, explaining, locating, planning, simulating.

Step 2 · Identify users and their tasks

Goal: 2 to 3 user profiles with 2 to 3 tasks each.

Users

User (role)	Expertise	Context (where, when, how often)
Database manager (secondary user)	Good at looking at trends, summaries, and data quality	Daily or weekly review of database health; uses overviews to find backlogs and quality problems.
Developer / Data Scientist (secondary/admin user)	Good at app usability and matching algorithms	Periodically when tuning similarity thresholds or checking model performance
Dietician (Primary User)	Strong knowledge of food composition, nutrition, and validation rules	Daily during validation of newly scanned products; often in short sessions at a desk/laptop or tablet; products arrive throughout the day.

User stories

Useful verbs: locate, compare, filter, drill down, explain, detect, simulate, export.
A story with only "see" or "view" is too vague. Ask: see it in order to do what?

#	User	As a [role], I want to [action in the app] so that I can [decision or outcome].
S1	Dietitian	**As a dietitian, I want to** compare newly scanned products side-by-side with suggested existing matches **so that I can** decide whether to merge them or approve a new item.
S2	Dietitian	**As a dietitian, I want to** filter unvalidated products by match confidence score **so that I can** clear clear-cut duplicates quickly and focus my time on difficult entries
S3	Dietitian	**As a dietitian, I want to** copy missing macronutrients from a verified active product with one click **so that I can** fix incomplete nutritional records without manual re-typing.
S4	Dietitian	**As a dietitian, I want to** reject incorrect duplicate suggestions **so that I can** validate genuinely new products into the database without losing unique items.
S5	Dietitian	**As a dietitian, I want to** search products by barcode or brand name **so that I can** manually inspect edge cases that the automated system missed.
S6	Database manager	**As a database manager, I want to** drill down into validated versus unvalidated product counts by food category **so that I can** identify where data backlogs are building up.
S7	Database manager	**As a database manager, I want to** export monthly reports on duplicate merge rates and data completeness **so that I can** track overall database quality improvements over time.
S8	Developer	**As a developer, I want to** simulate different similarity threshold settings **so that I can** optimize the matching algorithm and reduce false positives.
S9	Developer	**As a developer, I want to** track how often dietitians override match recommendations **so that I can** identify edge cases and retrain our text-matching models.

Step 3 · Derive the user questions

Goal: 4 to 8 numbered user questions. Each one is asked while working in the app and becomes at least one feature later.

Per user story: extract, break down (where? what does it look like? why? what if I act?), refine, review. If a question does not help the decision from Step 1, drop it.

> **Make the variable part explicit**
> **KPI:** "What was the booking rate in December?"
> **App:** "What was the booking rate for [selected customer group] during [selected period], compared to [baseline period]?"

Type: descriptive = what happened, where? · diagnostic = why? · predictive = what will happen? · What-if = what if I change X?

Acceptance criterion: one per question. Example: given a station is selected, when the user changes the date range, then the daily profile (Q2) updates.

#	User question, variable parts in [brackets]	Type	Acceptance criterion
Q1	How do newly scanned products compare to their top suggested matches based on name, category?	☑ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if **Story:** S1	**Given** an unvalidated product is selected, **When** the matching algorithm runs, **Then** a side-by-side comparison table displays the candidate products ranked by total similarity score with highlighted field differences.
Q2	Which unvalidated products fall within the **selected match confidence range**?	☑ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if **Story:** S2	**Given** the user sets a confidence filter slider (e.g., 85%–100%), **When** the filter is applied, **Then** the queue displays only unvalidated items meeting that score threshold.
Q3	Which micronutrients fields are missing from the **selected unvalidated product** and what values does it match?	☑ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if **Story:** S3	**Given** a candidate product is paired with an unvalidated item, **When** the user clicks "Fill Missing Data", **Then** empty macro fields on the target record are populated with values from the candidate record.
Q4	How do predicted yield, FP rate and manual review workload change if the **text weight** and **numeric weight** are adjusted?	☐ Descriptive ☐ Diagnostic ☐ Predictive ☑ What-if **Story:** S8	**Given** the Developer adjusts the text and numeric weight sliders in the simulator **When** the simulation is executed **Then** the dashboard displays predicted yield, false-positive estimates, and manual review workload.

Q5		☐ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S	**Given** **When** **Then**
Q6		☐ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S	**Given** **When** **Then**
Q7		☐ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S	**Given** **When** **Then**
Q8		☐ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S	**Given** **When** **Then**